



Harsh Raj Jaiswal

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● ABOUT ME

Aerospace Engineering graduate with strong academic and project-based experience in propulsion systems, aerodynamics, and computational fluid dynamics. Experienced in nozzle flow analysis, spacecraft conceptual design, and UAV performance evaluation, and motivated to pursue advanced studies in aerospace engineering with a focus on propulsion, CFD, and space systems.

● WORK EXPERIENCE

ENGINEERING INTERN – NEPAL AIRLINES CORPORATION – Jan 2025 – Jun 2025 – KATHMANDU, NEPAL

- Engineering internship supporting aircraft maintenance and airline engineering operations
- Assisted with technical documentation, maintenance records, and compliance procedures
- Observed aircraft systems inspection, routine maintenance tasks, and safety practices
- Gained exposure to operational workflows within an airline engineering environment

● EDUCATION AND TRAINING

AUG 2021 – JUN 2025 Punjab, India

BACHELOR OF ENGINEERING IN AEROSPACE ENGINEERING Chandigarh University

- Studied core aerospace subjects, including Aerodynamics, Aircraft Structures, Propulsion, Flight Mechanics, and Spacecraft Systems
- Developed strong analytical and simulation skills through academic projects and coursework

Website <https://www.cuchd.in/> | **Final grade** CGPA: 7.83/10

● PROJECTS

Design and Cold Flow Analysis of Aerospike Nozzle

- Conducted CFD-based cold flow simulations to study altitude-compensating nozzle behaviour
- Analysed pressure distribution, velocity contours, and thrust characteristics

Design and Performance Optimization of Dual-Bell Nozzle

- Studied flow separation and transition behaviour at varying pressure ratios
- Evaluated thrust efficiency and nozzle performance using numerical analysis

Spacecraft Design Project

- Performed conceptual spacecraft design, including mission analysis and subsystem sizing
- Worked on mass budgeting, propulsion selection, and configuration trade-offs

Solar-Powered UAV

- Designed a lightweight UAV focused on endurance and energy efficiency
- Performed propulsion sizing and preliminary aerodynamic performance estimation

● **SKILLS**

Aerospace Engineering Skills

Aerospace Propulsion System and Nozzle Design | Aerodynamics and Flight Mechanics

Simulation & Analysis Tools

Computational Fluid Dynamics (CFD) | Ansys Fluent

Programming & Computation

MATLAB for numerical analysis and simulations | Python for engineering applications

Design Tools

SolidWorks | CATIA

● **LANGUAGE SKILLS**

Mother tongue(s): **NEPALI** | **HINDI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1